

VIEW FROM PRACTICE

Frugal information systems (IS)

Richard T. Watson^{a*}, K. Niki Kunene^b and M. Sirajul Islam^c

^a*Terry College of Business, University of Georgia, Athens, GA, USA;* ^b*College of Business, University of Louisville, Louisville, KY, USA;* ^c*Swedish School of Business, Örebro University, Örebro, Sweden*

A discussion of the factors contributing to a likely increase in austerity establishes the case for a greater emphasis on frugal information systems (IS), which is defined as an information system that is developed and deployed with minimal resources to meet the pre-eminent goal of the client. The U-constructs are adopted as a foundation for a frugal IS, and two simple messaging system-based case studies illustrate their application. Design guidelines and practices for frugal IS are considered. The challenges and potential gains of frugal IS are reviewed.

Keywords: Frugal IS; IS strategies for development (national and sectoral); IS policy; sustainable development in developing and transition economies; development issues

The coming age of frugality

Frugality, being sparing or economical with regard to resources, is an appropriate philosophy for a world facing a shortage of resources, an increasing frequency of severe weather events, financial austerity in many developed economies, and where the majority of the population lives in highly resource-constrained economies. If the seven billion people currently on the globe were to have a Western European lifestyle, we would need 3.5 earths (Pimentel et al., 1999). Any shift toward some equitable sharing of the world's finite resources will require considerable frugality from the few wealthy to enable the many poor to have an improved lifestyle. Weather and human caused disasters damage and stress communication systems, and in such situations there is often a need to revert to frugal information systems (IS) because of the limited communication and computing resources. The large public and consumer debt of many advanced economies is impelling cuts in government and personal spending and introducing an era of financial restraint in many areas. As it is most unlikely that there will be a mass shift in current global consumption patterns, many countries will continue to operate in a highly resource-constrained environment.

While the earth is clearly overpopulated (Butler, 2004), we have a shortage of appropriately skilled human resources. China, with more than a billion people, exemplifies the talent shortage problem. Multinational companies operating in China, according to McKinsey & Co, find that few graduates of China's myriad universities have the skills to support a service-oriented business (Farrell & Grant, 2005). The greatest impediment to health in sub-Saharan Africa is the lack of health-care professionals, and it is estimated that there is a global shortage of 2.4 million doctors, nurses, and midwives (Anyangwe & Mtonga, 2007). This problem is exacerbated by phenomenon such as the African diaspora, whereby Africa's most talented and educated head for a better life in Europe and North America (Negash, Watson, & Straub, 2008). For example, more than 23% of the USA's nearly 800,000 physicians received their medical

*Corresponding author. Email: rwatson@terry.uga.edu

training outside the USA, the majority (64%) in low-income or lower middle-income countries (Hagopian, Thompson, Fordyce, Johnson, & Hart, 2004), those nations with typically the greatest need for physicians. The high-income societies, facing diminished natural resources and excessive debt, and the low-income societies, continuing to suffer from a severe shortage of nearly all resources, need to embrace frugality as a key design parameter for products and services. As IS are integral to the production and delivery of many products and services, it is timely to consider the notion of frugal IS.¹

Frugal engineering, the progenitor of frugal IS, has its origins in Tata's efforts to create a car, the Nano, for the Indian mass market. Ratan Tata, chairman of Tata Motors, set a goal of selling a car for INR 1 lakh (100,000) or about USD 2200 (Chang, 2008). While the engineers achieved the price target, there are problems with marketing what one Indian automotive engineer regards as a "brilliant concept" (*Economist*, 2011). Carlos Ghosn, the CEO of Renault-Nissan, is reputed to have coined the term "frugal engineering" to describe Tata's achievement in designing and manufacturing the Nano to a price considerably below current market. He replicated the approach in his firm's Romanian plant (*Economist*, 2011).

Frugal engineering has been extended to other products and services, such as a cell phone that is purely a phone, a handheld ultrasound scanner, and portable refrigerator the size of a small cooler. The advocates of frugal engineering advocate a "clean sheet" approach and the avoidance of unnecessary costs during design. An overriding design goal is to maximize the value to emerging economy customers who are cautious consumers because they have little to spend (Sehgal, Dehoff, & Panneer, 2010). Given our prognosis for a world that has to embrace frugality, we think of frugal engineering as jointly concerned with designing to minimize environmental impact and maximizing customer value. Of course, such diverse dual goals will create many trade-off issues, but that will be the reality of a world dominated by the drive for a sustainable civilization.

Definition of frugal IS

Now that we have some understanding of the circumstances promoting the emergence of frugal IS, we propose a definition. *A frugal IS is an information system that is developed and deployed with minimal resources to meet the pre-eminent goal of the client.* There are two key points in this definition. First, minimal resources are used. The typical project is managed within the triple constraints of time, resources, and scope. With frugal IS, these constraints are set at a very low level. Second, the prime goal of the client must be achieved. A system, frugal or not, must meet the goals of the client. The difference with frugal IS is that a system will typically have a single prime design goal, and this constrained scope will preclude consideration of secondary goals.

Foundations of frugal IS

We have adopted the four information drives as a model for exploring the dimensions of frugal IS. The drives have their origin as the U-construct in a marketing article describing the influence of IS on the future of commerce (Watson, Pitt, Berthon, & Zinkhan, 2002). This work was further extended to promote the idea that the four constructs are fundamental to understanding what consumers expect from IS (Junglas & Watson, 2006), and were relabeled as the four information drives (Table 1). The drives subsequently form the framework of a paper explaining the success of four mass transit systems designed to reduce carbon emissions (Watson, Boudreau, Chen, & Sepúlveda, 2011).

The success of an IS, including frugal systems and those serving people in emerging economies, is dependent on how well it satisfies the four information drives (Junglas & Watson,

Table 1. The information drives (Junglas & Watson, 2006).

Drive	Definition
Ubiquity	The drive to access information unconstrained by time and space
Uniqueness	The drive to know precisely the characteristics and location of a person or entity
Unison	The drive for information consistency
Universality	The drive to overcome the friction of IS's incompatibilities

2006). A frugal system should still strive to meet these drives, even though some might not be met as fully when resources are constrained. For example, the frugally engineered Nano has four wheels, an engine, and other necessary elements to meet the fundamental design goal of enabling a family to travel reliably. If we do not recognize the latent drives that have implicitly driven the development of IS for centuries, we might end up with the IS equivalent of a car without an engine. We save costs, but fail to fulfill basic needs. We now discuss each of these drives and the implications for frugal IS.

Ubiquity

Ubiquity is defined as the drive to have “access to information unconstrained by time and space” (Junglas & Watson, 2006). Most people, irrespective of where they live, now expect to access information wherever they might be.² Those with a rudimentary phone expect to be able to make phone calls and send and receive text messages. Consumers with smart phones expect to be able to surf the Internet, run apps to check flight delays, and so on. A frugal IS must support the ubiquity drive, but it might limit the amount of data transferred or the presentation format. The fundamental drive to access information should be met, even in a constrained manner, because it creates value by satisfying a key drive. Consequently, the requirements phase must determine what selected information should be delivered ubiquitously. For example, fishermen in India can get market prices for fish via their cell phones while at sea. They use this information to determine in which market to sell their fish to maximize their profits (Jensen, 2007).

Uniqueness

Uniqueness is the drive directed at “knowing precisely the characteristics and location of a person or entity” (Junglas & Watson, 2006). “Where am I and what is this?” are not new questions. Maps have existed for centuries to help people know where they are and to locate and identify objects within their environment. We have long been concerned with knowing about the people with whom we interact. For example, the early trader wanted to know if the buyer could pay his debt. The drive for uniqueness has spawned global positioning systems (GPS), barcodes, and radio frequency identification (RFID) to answer this fundamental drive in today's world. A frugal IS also needs to consider how the drive for uniqueness should be answered, even if done so in a rudimentary and constrained manner and at the same time creating considerable value for the information receiver.

Unison

In an information drive sense, unison is “information consistency” (Junglas & Watson, 2006). This includes ideas such as a single view of the customer and synchronization of calendars across devices. Conflicting information can result in confusion, indecision, and time wasted resolving the differences. It is a problem that bedevils the siloed organization or government.

More importantly, consumers and citizens receive lower quality service because they do not know which unit to deal with in an organization or government, and they often get different answers from different branches. Consumer and citizens do not envisage an organization or government as multiple units, rather they want it to operate as one unit with one public interface that can provide consistent and rapid responses to their questions. A frugal IS has a uniform (and ubiquitous) point of contact, because this will waste less consumer and citizen time, and a common or shared database because it is generally cheaper to establish and maintain one database rather than multiple databases. In the drive for unison, as expressed through frugal IS, subject? should result in better customer service at lower cost.

Universality

Universality describes the desire to “overcome the friction of information systems incompatibilities” (Junglas & Watson, 2006), and this describes the advent of standards (e.g. Open Document Format), currency unions (e.g. the euro), and development of the multi-functional smart phones (e.g. phone, GPS, camera, personal digital assistant, and media player). A frugal IS must be compatible with existing systems so that it does not add to the incompatibility problem. This means using existing standards, and preferably open standards as these lower costs, and existing objects that the target audience is likely to already own. For example, many people have cell phones, so these are an obvious delivery platform for information services. In some countries, smart cards are used extensively in mass transit systems and by banks, and many people have them. As a result, they can become the foundation for handling other low-value transactions. Frugal IS reuses the existing infrastructure to create new systems.

Infomimicry

Humans are not the only species that processes information. Indeed we are rather latecomers to the scene. Human information processing, with its origins in those species that preceded *homo sapiens*, is a few million years old. We might be able to learn from other species, which while having less-developed IS than humans, have effective systems. Biomimicry (Benyus, 2002) studies the design and function of biological entities and processes to extract lessons from millions of years of evolution. Thus, we can define infomimicry as the study of the design and function of the IS of other species to gain insights from eons of evolution. In our case, we think infomimicry might have considerable potential for frugal IS because non-human information systems are far less elaborate. For example, ants, the product of 140 million years of evolution, use pheromones to exchange information for tasks such as identifying a food source. Pheromones are hormones secreted by one ant and smelled or tasted by members of the same colony. A particular species of ant will emit between 10 and 20 different pheromones, and members of that species are able to interpret instinctually the meaning of each pheromone. Some scholars are applying knowledge gained from studying ants' IS to grid computing (e.g. Xu, Hou, & Sun, 2003). Imagine that you were to set a design goal of creating a communication system limited to 10–20 symbols to support disasters, could infomimicry inform design? If interest develops in frugal IS, then we need scholars to also develop the field of infomimicry so that IS can benefit from evolution over millions of years that has created effective information systems designs.

Design guidelines and practices

We now provide a set of principles and good practices that we think will effectively guide the design of a frugal IS.³ We indicate where one or more of the information drives fits a particular principle or practice.

Be frugal with time, resources, and scope

Set mind-changing low upper limits and force systems designers to rethink what they can achieve and how they might do it. Design for low cost rather than taking cost out of the design at a latter stage.

Set open source software as the default choice

Open source software is typically free and more likely to conform to open standards (universality) than proprietary software. Given the limited nature of a frugal IS, there may be no need for a service contract to support open source software, particularly as a frugal IS should have limited support requirements.

Public rather than proprietary data

In an information economy, many companies create a business around selling data (e.g. credit reporting agencies), but public agencies (such as governments, the UN, and World Bank) are also data sources. In the first instance, the frugal IS designer should attempt to deliver value based on free public data. For example, the public weather forecasting system might provide sufficient accuracy and detail for most citizens, whereas a proprietary service adds the precision required for commercial operations.

Use something the client already has

There are considerable software savings when the client pays for a share of the infrastructure. The early pioneers of Internet banking, for example, found that because their customers had a microcomputer and Internet connection, their cost per transaction was USD 0.01 compared with an ATM at USD 0.27 and a bank teller at USD 1.08 (Rothstein & Watson, 2004). In the developed and developing economies, most clients have a cell phone (universality), and this can be the platform for delivering a frugal IS. Furthermore, a frugal IS should integrate with, where appropriate, existing apps people are likely to already have on their cell phones (universality) to minimize information friction.

Simplify processes first

Following the principles of business process re-engineering (Hammer & Champy, 1993), first simplify existing processes so that the frugal IS can be even more frugal. Simplification should also raise the levels of unison and universality.

Crowdsource for innovation

Networks have made feasible the idea of getting many people to work on a single problem to accelerate innovation and unearth breakthrough ideas. Crowdsourcing (Brabham, 2008) should be an early step in frugal IS design to reveal potentially novel and low-cost ways of providing information in a highly constrained environment. It could also be used to explore ways to take costs and complexity out of existing or proposed systems.

Use interns, students, or those with experience in a frugal setting

Companies who hire interns or support student projects could consider assigning frugal IS work to these groups. This lowers costs. Furthermore, their limited knowledge could be an advantage,

as they will tend to be less encumbered by the implicit assumptions that can accompany experience. Alternatively, assign employees who have some experience in working in resource-constrained settings. An Australian engineering firm recruits engineers from Africa because they are far more cost conscious when designing solutions than engineers educated in affluent Australia.⁴

Learn from frugal engineering

Frugal engineering has a few years' start on frugal IS, so we should be able to learn from it. The knowledge gained by engineers working on the Nano and other frugal projects could be extracted and transferred to an IS setting. We probably need one or two case studies on frugal engineering so that the lessons are more accessible. Of course, we also need to encourage frugal IS cases to be written, and this article is a step in that direction.

Celebrate frugal solutions

Encourage a culture of frugality by highlighting system successes that demonstrate frugal IS principles in action. What gets rewarded and celebrated sends a strong signal to other employees.

Do it quickly

Set relatively tight deadlines, and aim to “deliver” the system sooner rather than later. Leveraging time constraints can have a positive effect on cost. In other words, be frugal with time.

Two cases studies illustrating frugal IS in action

Two cases illustrate the development and deployment of simple messaging system (SMS)-based frugal IS to improve public knowledge of crisis situations (Ushahidi.com) and to give Bangladeshi farmers better information about market prices (Agriculture Market Information Service, AMIS). We follow the description of each case with an illustration of how it satisfies the four U-constructs.

Ushahidi.com

Ushahidi was conceived originally as a way to gather information from the public via the web and mobile phones in crisis situations. In December 2007, Kenya was in the midst of post-election ethnic violence between the supporters of President Kibaki and his opponents. The government had banned mainstream media from reporting on the disorder, and it ceased broadcasting any news related to the strife. A Kenyan political activist, Oryō Okolloh, blogged about the situation and invited readers to comment on the violence they were witnessing as a way of getting reliable information. Comments came flooding in, and Okolloh would manually aggregate these reports. Within a couple of days, however, Okolloh could no longer keep up with the information flow, so she challenged “the techies” who followed her blog to help solve this problem. She envisioned a service that would automatically aggregate citizen reports as well as provide near real-time mapping to locate the reported attacks (Shirky, 2011). Two Kenyan programmers, Erik “Hash” Hersman and David Kobia, responded (Komo, 2008) and via a conference call the three hashed out the basic system requirements> Within three days the first beta version of Ushahidi went live (Komo, 2008; Shirky, 2011).

Very quickly a system was developed using the web and text-messaging to gather information and a geo-mapping engine to visualize it. Hersman acknowledged later that shortcuts

were made in creating the geo-mapping engine as the violence was spreading rapidly in January (Komo, 2008). Text messages flooded in. Ushahidi had over 45,000 visitors in Kenya alone (Ushahidi.com, 2011). Usage evolved from first-person accounts of the violence to support for peace efforts.

When anti-immigrant violence erupted a few months later in South Africa, the system was ported to create UnitedforAfrica.co.za (Komo, 2008; Talbot, 2008). The early success prompted the developers to develop Ushahidi into a platform where anyone could set up a service for collecting and mapping information reported via text messaging (Talbot, 2008). Ushahidi is now an “open source software for information collection, visualization, and interactive mapping” (Ushahidi.com, 2011), and has since been used in the Democratic Republic of Congo (violence), Haiti (earthquake), New Zealand (earthquake), and Japan (earthquake).

We can understand the success of Ushahidi by viewing it through the U-concepts.

Ubiquity

Ushahidi is a very good exemplar of a system designed with ubiquity as a central premise. The service was initially implemented to be accessed via the web. The developers quickly added a capacity to interface with the systems via mobile phone text messaging, thus piggybacking on the ubiquity of mobile telephony.

Uniqueness

A study comparing Ushahidi, citizen journalists, and mainstream media found that “contrary to news media and citizen journalists, Ushahidi data always had specific location information” (Meier & Brodock, 2008). Ushahidi supports uniqueness, because it can be used to log the nature of an incident and its location.

Unison

For data collection, the original Ushahidi website interface used colored push pin icons to represent acts of violence: for example, blue was for riots, red for deaths, and a fire icon for all incidents (Komo, 2008). These symbols were comprehensive enough for the task, consistent, and easy for users to understand. For data presentation, geo-mapping to Google maps provided a consistently rich medium of layered data.

Universality

Ushahidi is compatible with the SMS found on mobile phones. As a result, people with a cell phone can be a node on the Ushahidi network. By using a universally available technology, Ushahidi made everyone a potential reporter of violence.

A mobile phone-based amisAMIS

With a population of more than 160 million, Bangladesh is one of the least developed and most highly densely populated countries of the world. Farmers are the backbone of this country. They contribute to the food security of millions of people and are 62% of the total workforce. Unfortunately, despite their inevitable roles in the national economy in general and rural development in particular, they are, as a community, neglected in terms of economic development and social justice. They are truly at the bottom of the pyramid. According to the World Bank, “Poverty in Bangladesh is primarily a “rural phenomenon, with 53 percent of its rural population classified

as poor, comprising about 85 percent of the country's poor" (World Bank, 2011). Agriculture in Bangladesh contributes only 13% to gross domestic product (Bangladesh Bureau of Statistics, 2006). Furthermore, "growth in agriculture or in related activities produces a stronger impact on poverty, mainly because most of the people live and work in those sectors" (Danielson, 2001, p. 17). Increased employment in agriculture can influence the growth of this sector. However, records show that agriculture employment in Bangladesh is declining (Bangladesh Bureau of Statistics, 2002). One key cause is that farmers are becoming dissatisfied with farming. They do not get a reasonable return on their investment in land and labor. There is a huge disparity in the prices of agricultural produce (particularly for perishable commodities) between the rural growers and urban retailers. Due to superior market knowledge, the middlemen, in most cases, gain higher spatial as well as temporal arbitrage by exploiting less knowledgeable farmers.

There is a need to provide farmers with better access to market information (Islam, 2011; Jensen, 2007) so that they can gain bargaining power with the middlemen as well as to take rational decisions when selling the commodities resulting from their labor. As part of a solution, the Ministry of Agriculture of the Government of Bangladesh initiated a web-based market information service (www.dam.gov.bd). However, this has so far been found to be an inappropriate, ineffective solution, in the rural context of Bangladesh, as most farmers do not have access to the Internet and the presence of such a service is unknown to most of them (Islam & Grönlund, 2007). In addition, the information content of this service is not sensitive to individual preferences and the quality of the information in terms of time, place, and formats is not reliable.

A widely and easily available mobile phone-based AMIS capable of providing quality information with the participation of the local level actors should be a better technological choice (Islam & Grönlund, 2007). A mobile AMIS, which was locally branded as *PalliNet* or RuralNet (Islam, 2011), was introduced in early 2009 on a test basis at some remote villages in a northern district of Bangladesh. The system was designed and developed with continuous feedback from the targeted clients – the farmers – and based on the existing telecommunications infrastructure. While *PalliNet* was accessible by anyone in Bangladesh during the test period, a pilot study was conducted with 100 registered farmers (divided into "push" and "pull" groups), who had mobile phones (mostly Nokia's ultra-basic 1000 series). These farmers were chosen from those villages connected to three neighboring wholesale and retail markets. Three price collectors from these three markets were appointed. They gather maximum wholesale and retail price information of a certain list of agricultural commodities from their respective markets every day during the early trading hours and send the collected information to the server via SMS in a prescribed format. The server then disseminates the price information via SMS in a native phonetic (Bangla) to the registered farmers for no charge within business hours on the same day according to their predefined preferences (i.e. preferred crops and markets). This timely and reliable market information helps farmers to make rational decisions on selling their produce. The post-implementation survey revealed (Islam & Grönlund, 2010) that the farmers using *PalliNet* gained infopowerment, a sense of information-based empowerment that ameliorated their previous lack of market information and power to bargain.

The *PalliNet* case can be evaluated from the perspectives of U-concepts as follows:

Ubiquity

Access to *PalliNet* is unconstrained by time and place, as the system is based on mobile phone technology. Overcoming these constraints is not possible with the Government's web-based AMIS as most of the farmers do not have easy and affordable access to the Internet. Furthermore, farmers' general level of literacy and the technological limitation of their mobile handsets are not appropriate for a web-based AMIS.

Uniqueness

PalliNet is capable of identifying individual farmers from their cell numbers and thus meeting their preferences. The cell numbers of the respective price collectors are predefined in the system. Thus, the system knows who is who, what information to accept, what requests to deny, and what information to supply on an individual basis.

Unison

In *PalliNet*, there are numerical codes for each of the markets and for each of the crops available within a particular region. The system accepts price data only from registered collectors. Furthermore, it generates an error message if data elements are received in the wrong format and provides acknowledgement messages for all data successfully received in the right format. There is no possibility of conflicting information because one price collector is responsible for one particular market. In addition, a local level manager is responsible for overseeing the three price collectors to ensure data quality.

Universality

PalliNet is compatible with the basic features of the mobile phones available to the farmers, using SMS. The contents of each SMS message cover most of the required information within the limitation of 160 characters in a single SMS. *PalliNet* is based on a contextually embedded technology that leverages a standard platform that was already owned by the farming community. It takes advantage of low-cost data management procedures and disseminates quality information in accordance with individual preferences.

The challenges of frugal IS

Limited resources are not the sole challenges when designing a frugal IS. It might require considerable intellectual effort to break away from traditional systems development protocols and harness creativity to make a cost breakthrough. We become accustomed to thinking about solving problems in a particular way and using certain technologies. Frugal IS design will likely require a more expansive and open approach to design alternatives.

Frugal design does not imply low complexity. We might think of a basic cell phone as simple, but a highly complex telecommunications network supports it. A frugal IS might well sit on top of an intricate infrastructure, but this complexity needs to be hidden and it needs to contribute to the frugality of the solution. Complex network structures, such as the Internet and cell phone system, can provide basic services at low cost because their immense scalability has lowered the cost of usage over time. Frugal IS designers find ways to build simplicity on top of complexity.

Because the design of a frugal IS is likely to be based on an existing infrastructure, it is dependent on the long-term viability of the business model of the supporting system. Inherently, this is no different from selecting a platform for any IS (e.g. the OS or middleware). Given, however, that we expect people at the bottom of the pyramid to be major beneficiaries of a frugal IS, the consequences of a platform failure could be quite severe for a large portion of the population. For example, the Indian fisherman who could no longer get current market prices via his cell phone would likely suffer reduced income.

Frugality has its savings, but it also has costs in terms of opportunities forgone. In designing a frugal IS, the designers should consider what incremental value might be gained from providing additional information, which might become features of a later release. However,

due caution needs to be exercised so that the benefits of frugality are not lost to system escalation. Nearly every system has the Oliver Twist syndrome of “Please sir, I want some more,” and a frugal IS should generally err on the side of denying such requests.

The potential gains of frugal IS

The divide between rich and poor, regionally and globally, is also an information divide. The rich have access to many sources of information, and their information drives are increasingly being met by innovations such as smart phones, cloud computing, 4G networks, and GPS. Fortunately, the number of rich people able to afford the technology for an information-intensive life is sufficiently large to provide the scaling effects that drive down the cost of achieving these same information drives in a limited fashion for the many less rich of this world. The poor cannot consume information at the same level as the rich, but by designing frugal IS we can provide them with the information that is critical to their quality of life. While there are other potential gains from frugal IS, such as highly reliable disaster communication systems, we think the greatest gain will come from infopowerment (information + empowerment) of those at the bottom of the pyramid.

Acknowledgements

We appreciate the comments of John C. Johnson of the International Monetary Fund and the anonymous reviewer.

Notes

1. We debated whether to use Frugal IS or frugal IS, and in the interests of frugality settled on frugal IS.
2. Global cell phone adoption is 87% and 79% in the developed and developing world, respectively (International Telecommunications Union, 2011).
3. The attendees at the Association for Information Systems SIG Global Development workshop at the Americas Conference on Information Systems in Detroit in July 2011 proposed many of these principles and practices.
4. Personal correspondence.

Notes on contributors

Richard Watson is the J. Rex Fuqua Distinguished Chair for Internet Strategy in the Terry College of Business at the University of Georgia. He has published over 150 journal articles, and written books on electronic commerce, data management and energy informatics. He has given invited presentations in more than 30 countries. He is a former President of the Association for Information Systems, a visiting professor at the Viktoria Institute in Sweden, the International Coordinator for the PhD in Information Systems at Addis Ababa University in Ethiopia, and the Research Director for the Advanced Practices Council of the Society of Information Management, an exclusive forum for senior IS executives. In 2011, he received the Association for Information Systems’ LEO award, which is given for exceptional life time achievement in Information Systems.

K. Niki Kunene is an assistant professor of Computer Information systems at the University of Louisville. She received her PhD from the Virginia Commonwealth University (2006). Her research interests are interdisciplinary and include information, communications technologies for development (ICT4D) health informatics, data mining, decision support systems, and design science research. Her work has appeared in the *European Journal of Operational Research*, and several medical journals including the *European Journal of Neurosurgery*.

M. Sirajul Islam is an Assistant Professor at the School of Business of Örebro University, Sweden. He received his PhD and MSc in Informatics from the Informatics department of Örebro University. He is

the recipient of the Börje Langefors best doctoral dissertation award in Sweden by Swedish Academy of IS (sis-net.se) for the year 2011. Dr. Islam is a founding President of the Bangladesh Centre for E-governance, elected fellow of the UK-based World Innovation Foundation and one of the recipients of the 'Strengthening ICTD Research Capacity in Asia (SIRCA)' fellowships awarded by Nanyang Technology University, Singapore. He has also served as a faculty member in several universities in Bangladesh. As a member of the ICT4D and eGovernment research groups at Örebro University, he coordinates the 'International Network for Postgraduate Students in the Area of ICT4D (IPID)', funded by SPIDER/SIDA.

References

- Anyangwe, S.C., & Mtonga, C. (2007). Inequities in the global health workforce: The greatest impediment to health in sub-Saharan Africa. *International Journal of Environmental Research and Public Health*, 4(2), 93–100, doi: 10.3390/ijerph2007040002.
- Bangladesh Bureau of Statistics. (2002). *Labor force survey Bangladesh 1999–2000* (Trans.).
- Bangladesh Bureau of Statistics. (2006). *Handbook of agricultural statistics* (Trans.). Retrieved from <http://www.moa.gov.bd/statistics/bag.htm>
- Benyus, J.M. (2002). *Biomimicry: Innovation inspired by nature*. New York: Harper Perennial.
- Brabham, D.C. (2008). Crowdsourcing as a model for problem solving. *Convergence: The International Journal of Research into New Media Technologies*, 14(1), 75, doi: 10.1177/1354856507084420.
- Butler, C.D. (2004). Human carrying capacity and human health. *PLoS Med*, 1(3), e55, doi: 10.1371/journal.pmed.0010055.
- Chang, R. (2008, January 10). Tata Nano: The world's cheapest car. *The New York Times*. Retrieved from <http://heels.blogs.nytimes.com/2008/01/10/tata-nano-the-worlds-cheapest-car/>.
- Danielson, A. (2001). *When do the poor benefit from growth, and why?* (Trans.). Retrieved from http://swopec.hhs.se/lunewp/papers/lunewp2001_012.pdf
- Economist*. (2011, August 20). Tata's Nano: Stuck in low gear. *Economist*. Retrieved from <http://www.economist.com/node/21526374>.
- Farrell, D., & Grant, A.J. (2005). China's looming talent shortage. *The McKinsey Quarterly*, 4(56), 70–79.
- Hagopian, A., Thompson, M.J., Fordyce, M., Johnson, K.E., & Hart, L.G. (2004). The migration of physicians from sub-Saharan Africa to the United States of America: Measures of the African brain drain. *Human Resources for Health*, 2(1), 17, doi: 10.1186/1478-4491-2-17.
- Hammer, M., & Champy, J. (1993). *Reengineering the corporation*. New York: Harper.
- International Telecommunications Union. (2011). *ICT facts and figures* (Trans.). Geneva, Switzerland. Retrieved from <http://www.itu.int/ITU-D/ict/facts/2011/material/ICTFactsFigures2011.pdf>
- Islam, M.S. (2011). *Creating opportunity by connecting the unconnected: Mobile phone based agriculture market information service for farmers in Bangladesh* (PhD, Örebro University). Retrieved from <http://oru.diva-portal.org/smash/record.jsf?pid=diva2:440147>
- Islam, M.S., & Grönlund, Å. (2007). *Agriculture market information e-service in Bangladesh: A stakeholder-oriented case analysis*. Paper presented at the 6th International Conference (EGOV 07), Regensburg.
- Islam, M.S., & Grönlund, A. (2010). An agricultural market information service (AMIS) in Bangladesh: Evaluating a mobile phone based e-service in a rural context. *Information Development*, 26(4), 289, doi: 10.1177/0266666910385556.
- Jensen, R. (2007). The digital divide: Information (technology), market performance, and welfare in the South Indian fisheries sector. *The Quarterly Journal of Economics*, 122(3), 879–924, doi: 10.1162/qjec.122.3.879.
- Junglas, I.A., & Watson, R.T. (2006). The U-constructs: Four information drives. *Communications of AIS*, 17, 569–592.
- Komo, D. (2008). Kenyan Techie honored for role in tracking post-election violence. *Mshale.com*. Retrieved from <http://mshale.com/article.cfm?articleID=18192>
- Meier, P., & Brodock, K. (2008). Crisis mapping Kenya's election violence. *Harvard Humanitarian Initiative, HHI*.
- Negash, S., Watson, R.T., & Straub, D. (2008). An African PhD program in information systems: The case of Addis Ababa University, Ethiopia. *The African Journal of Information Systems*, 1(1), 67–76.
- Pimentel, D., Bailey, O., Kim, P., Mullaney, E., Calabrese, J., Walman, L., & Yao, X. (1999). Will limits of the earth's resources control human numbers? *Environment, Development and Sustainability*, 1(1), 19–39.

- Rothstein, C., & Watson, R.T. (2004). NetBank: The conservative Internet entrepreneurs. *Communications of the AIS*, 14(10), 1–24.
- Sehgal, V., Dehoff, K., & Panneer, G. (May 25). The importance of Frugal engineering. *Strategy+Business*, Summer(59), 1–5.
- Shirky, C. (2011). *Cognitive surplus: How technology makes consumers into collaborators*. New York: Penguin Press.
- Talbot, D. (2008). Ten web startups to watch: Crisis sourcing. *Technologyreview.com* (July/August). Retrieved from <http://www.technologyreview.com/Biztech/20923/page5/>
- Watson, R.T., Boudreau, M.-C., Chen, A.J., & Sepúlveda, H.H. (2011). Green projects: An information drives analysis of four cases. *The Journal of Strategic Information Systems*, 20(1), 55–62, doi: 10.1016/j.jsis.2010.09.004.
- Watson, R.T., Pitt, L.F., Berthon, P., & Zinkhan, G.M. (2002). U-commerce: Expanding the universe of marketing. *Journal of the Academy of Marketing Science*, 30(4), 329–343, doi: 10.1177/009207002236909.
- World Bank. (2011). *Bangladesh: Priorities for agriculture and rural development*. Retrieved from <http://go.worldbank.org/770VR4DIU0>
- Xu, Z., Hou, X., & Sun, J. (2003). *Ant algorithm-based task scheduling in grid computing*. Paper presented at the Canadian Conference on Electrical and Computer Engineering, 2003., Montreal. Vol. 2, 1107–1110.

Copyright of Information Technology for Development is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.